Read me:

1. This file displays the phenotypes of **complemented lines for ARENA mutants and their immunoblotting results**. The catalog is on Slide 2.
2. A phenotype image of the original ARENA mutant is shown on each slide and labeled with mutant ID and gene information. See **Supplementary Table 2** for details.
3. Transformants derived from the mutant background with plasmids carrying wild-type genes were arrayed for phenotyping. Each array includes two colonies of wild-type GV32 (WT) and two of the original mutant as controls.
4. 'M' refers to the original mutant; 'Cx' to phenotypically-rescued lines; 'Nx' to transformants with mutant-like phenotypes.
5. For each gene, over 10% of transformants exhibited rescued phenotypes, explaining why many 'Cx' line neighbors also appeared wild-type.
6. Immunoblotting with an anti-FLAG antibody showed expected fusion protein bands in 'Cx' lines but not in 'Nx' lines for *PMR1*, *CPLD38*, and *CCS5*. Other cases had limited signal-to-background ratios, likely due to transgene expression challenges in *C. reinhardtii.* The high percentage of rescued phenotypes, beyond what random suppressor mutations would suggest, is the basis for our conclusion of successful genetic validation.
7. For each ARENA mutant, two or three lines have been cryopreserved for community sharing to facilitate further research.
8. In the immunoblotting results, the red arrows indicate the bands corresponding to the fusion protein, which are close in size to the expected band. Notably, in the immunoblot of the *hvfn1* complementary strains, we observed two bands.

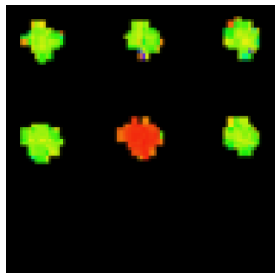
Catalog

Mutant ID	Mutant name	Slide or figure
arn0638	*pmr1*	Slide 3
arn0086	*rsga*	Slide 4
arn0415	*not11-1*	Slide 5
arn0586	*cpld38*	Slide 6
arn0095	*ccs5*	Slide 7
arn0525	*hvfn1*	Slide 8
arn0068	*tsp2*	Figure 3

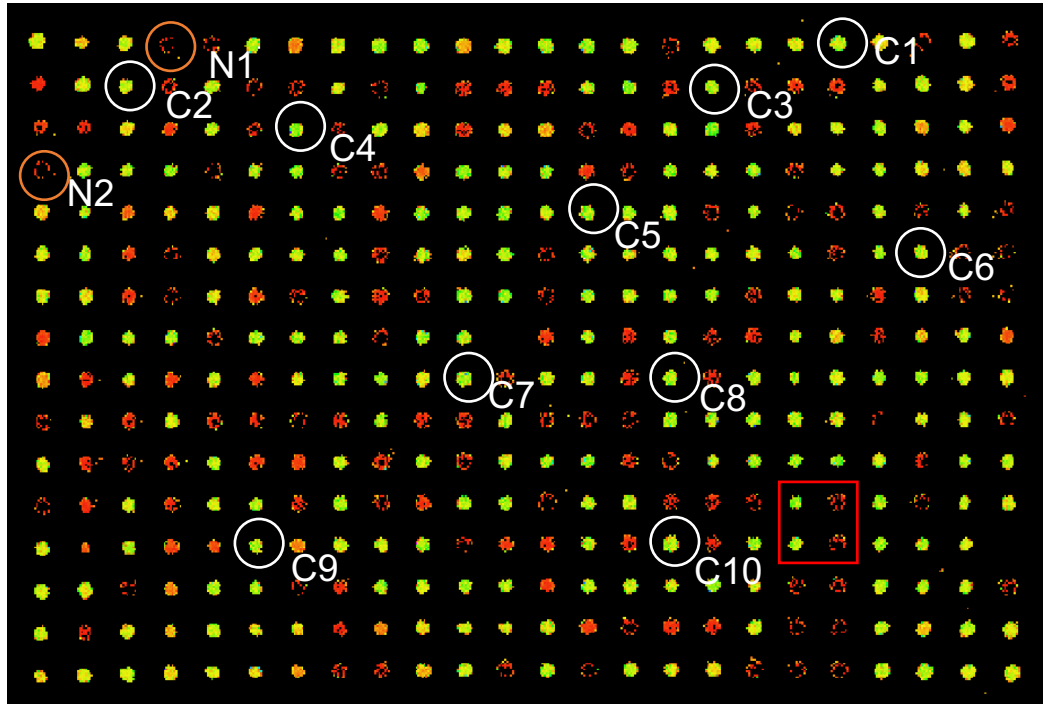
pmr1, arn0638

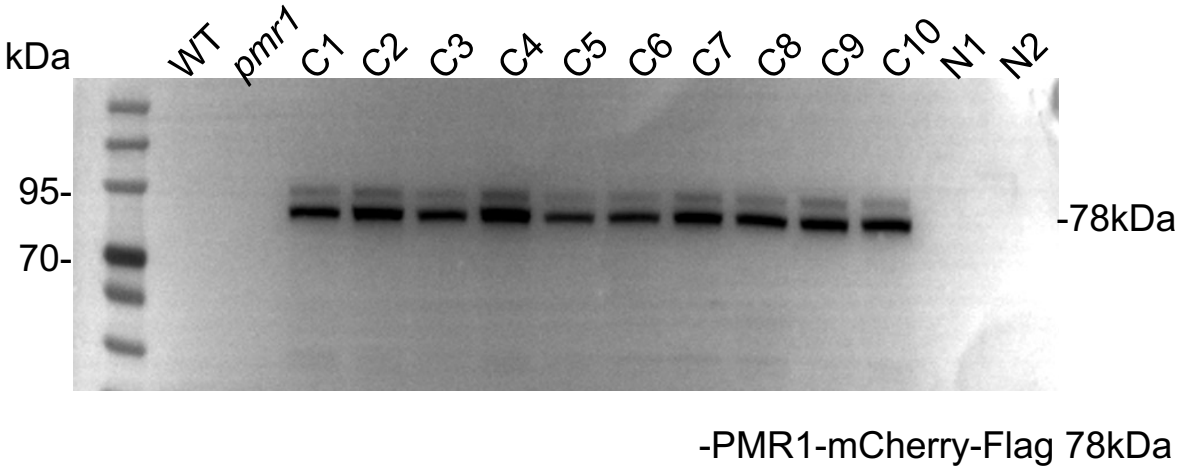
PMR1, Cre10.g448950

Nocturnin (CCRN4L)



−N_low NPQ





-PMR1-mCherry-Flag 78kDa

WT	m
WT	m

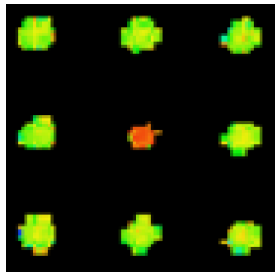


-PMR1-mCherry-FLAG 78 kDa

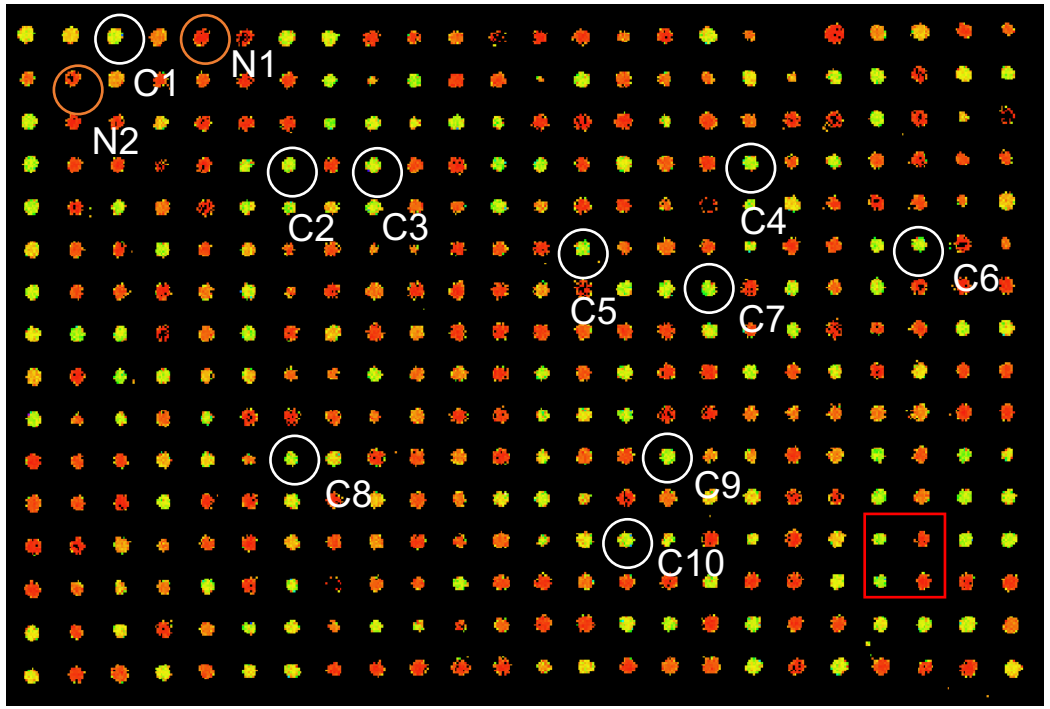
rsga, arn0086

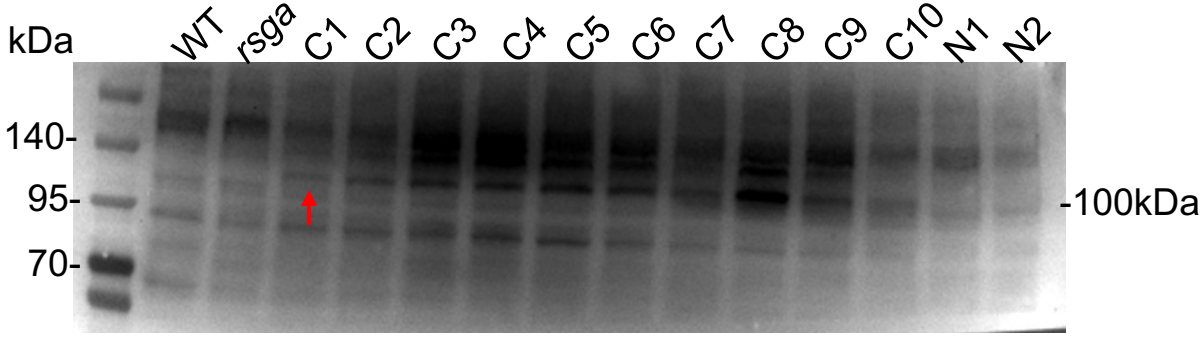
RSGA, Cre10.g436600

Ribosome biogenesis GTPase (rsgA, engC)

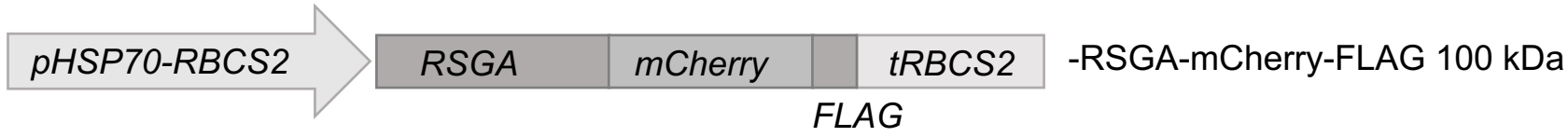


−N_low NPQ





WT	m
WT	m

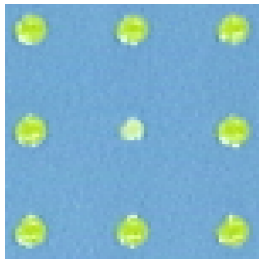


-RSGA-mCherry-FLAG 100 kDa

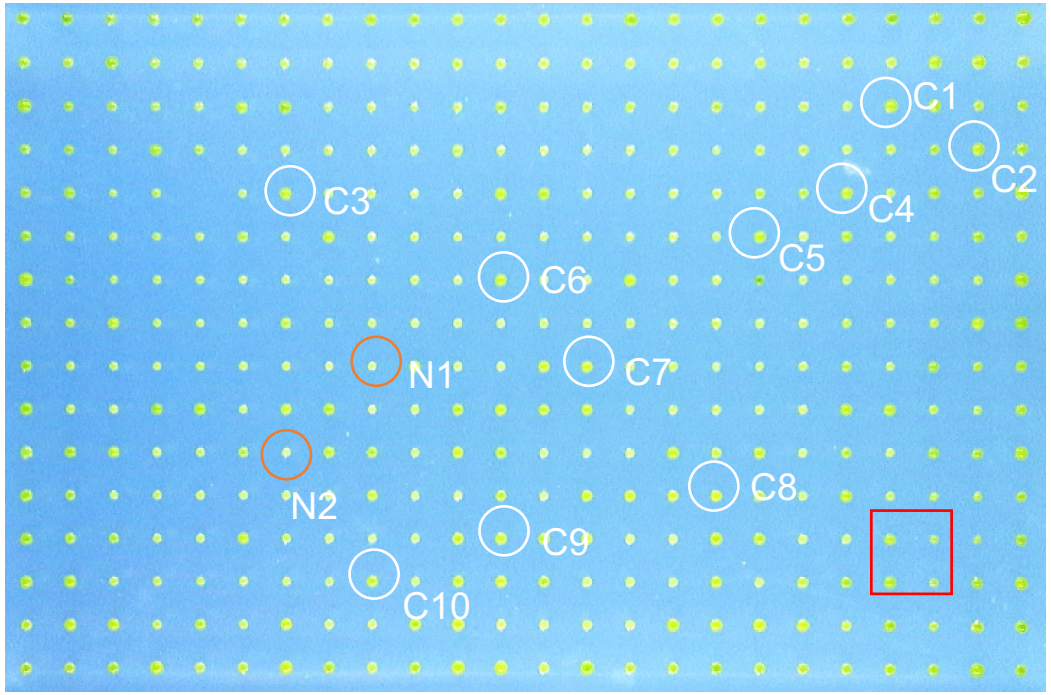
not11-1, arn0415

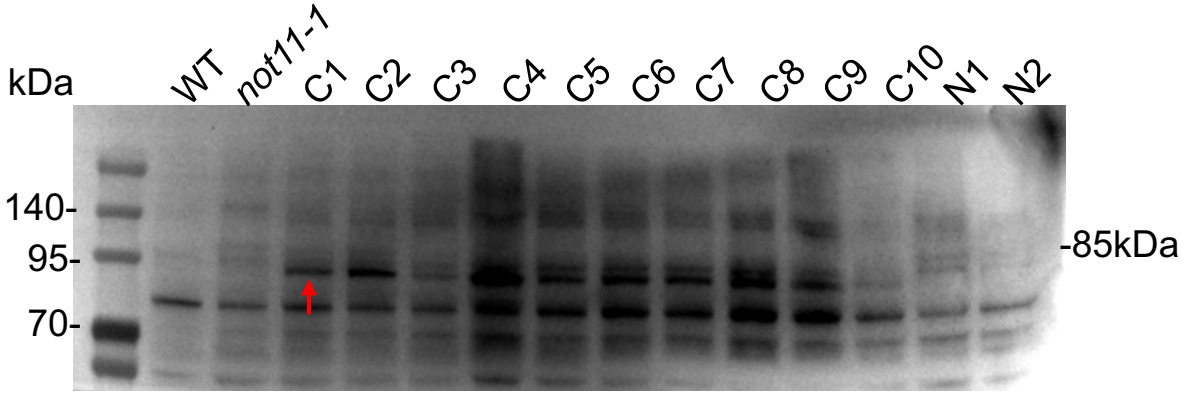
NOT11, Cre17.g726800

Uncharacterized conserved protein (DUF2363)



−N_fast chlorosis





WT m

WT m

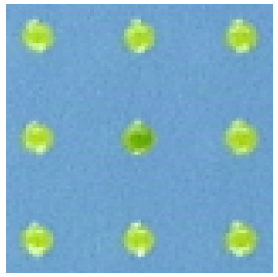


-NOT11-mCherry-FLAG 85 kDa

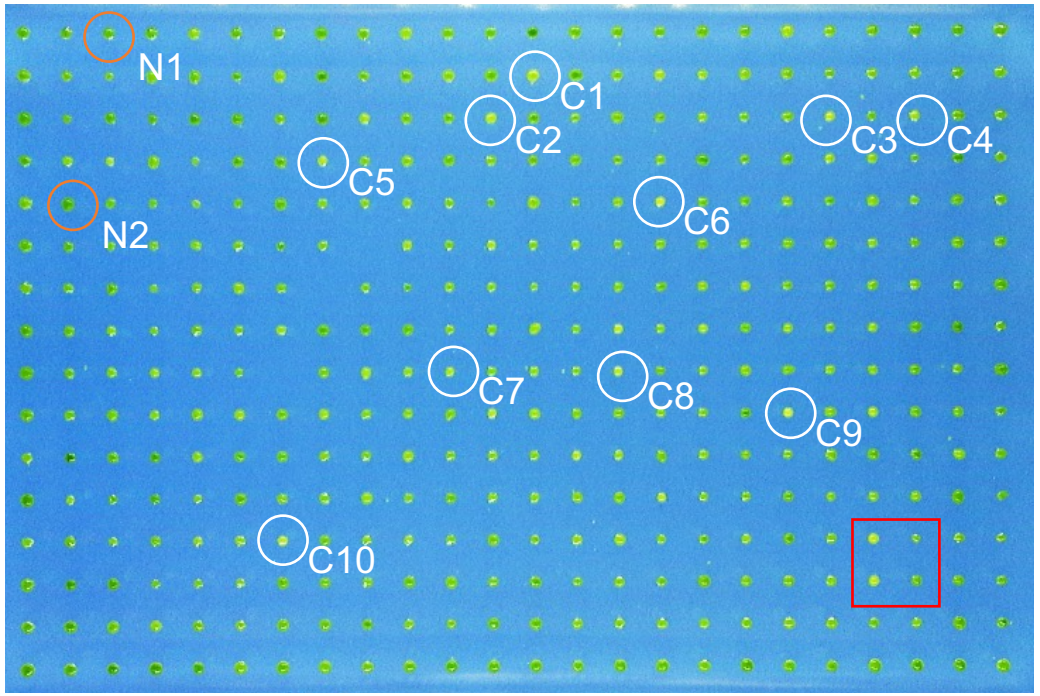
cpld38, arn0586

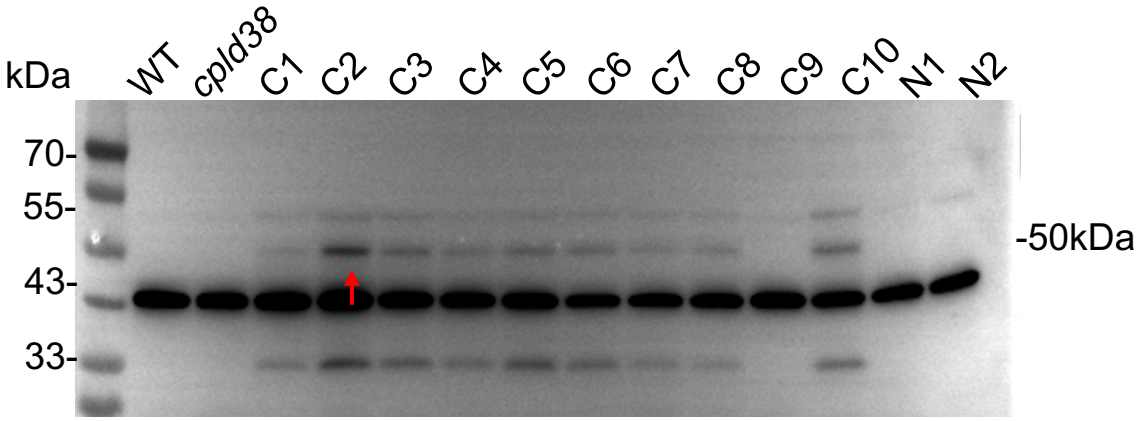
CPLD38, Cre01.g000850

Predicted protein, CPLD38



−N_delayed chlorosis





WT m

WT m

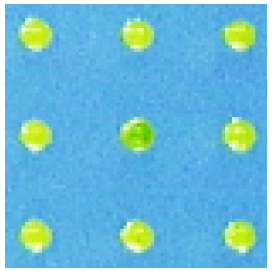


-CPLD38-mCherry-FLAG 50 kDa

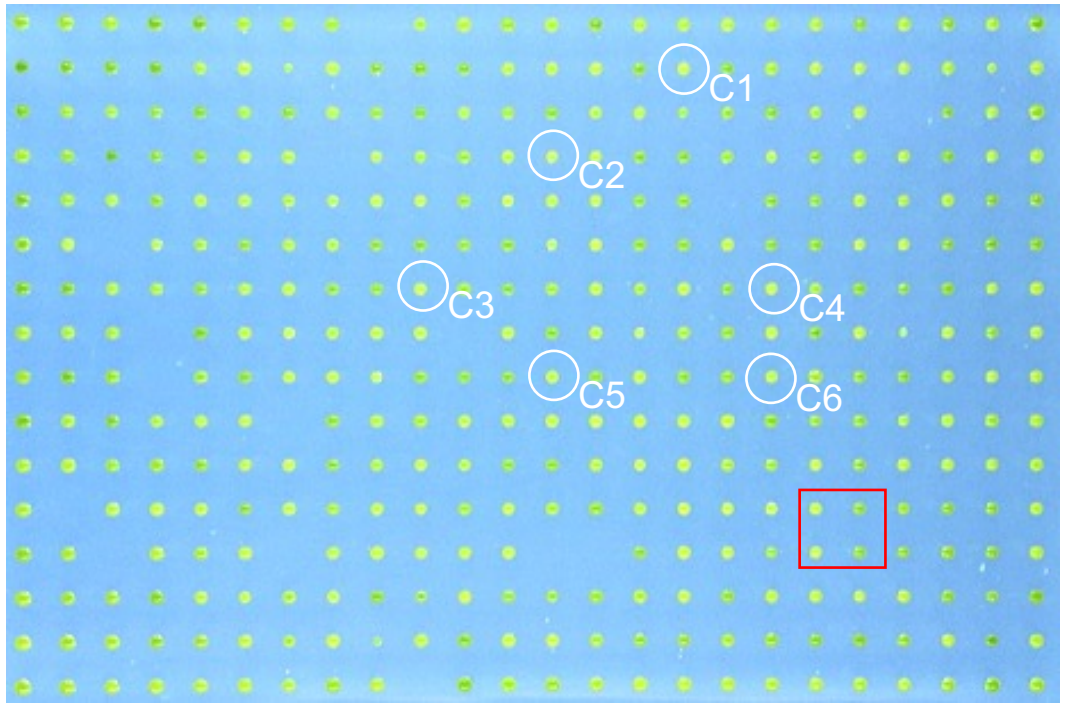
ccs5, arn0095

CCS5, Cre17.g702150

Thioredoxin-like protein hcf164, chloroplastic HCF164, TRX20, CCS5

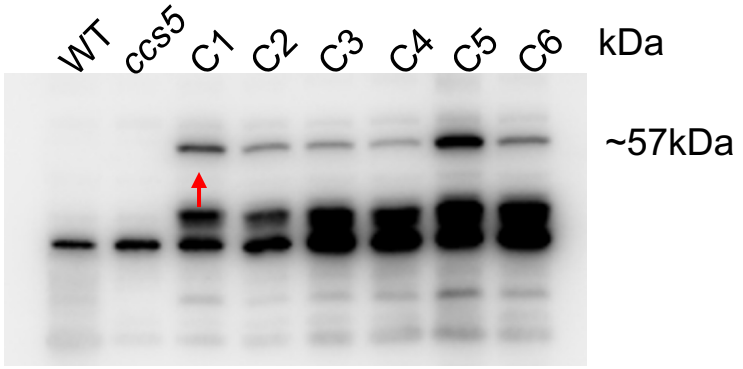


−N_delayed chlorosis



WT m

WT m



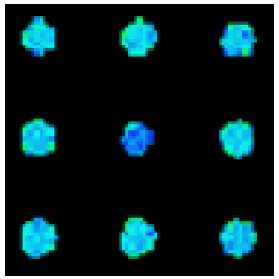


-CCS5-mCherry-FLAG 57 kDa

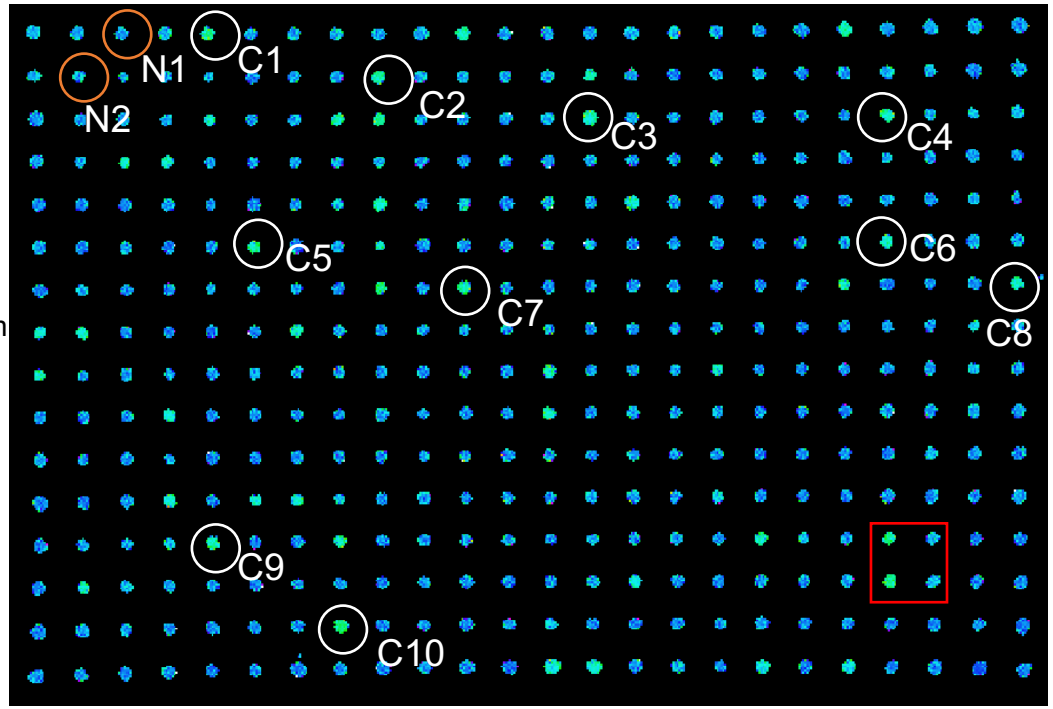
hvfn1, arn0525

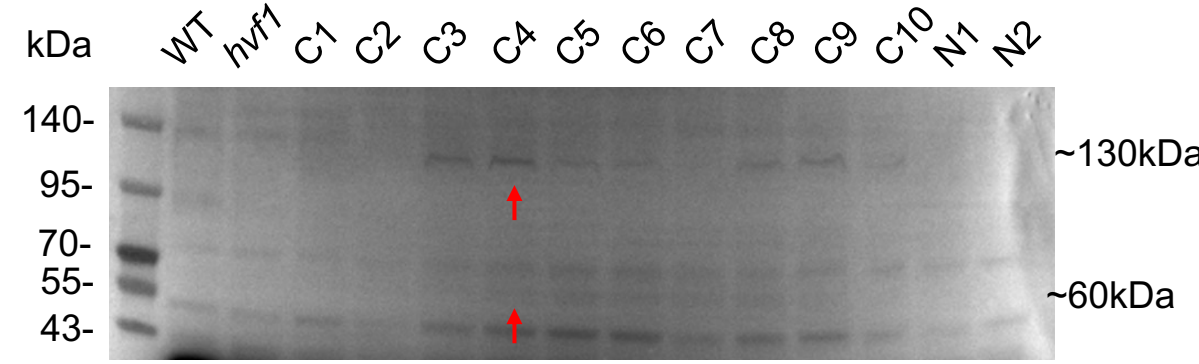
HVFN1, Cre10.g424550

Coexpressed with genes in phototrophic condition specific coexpression subnetwork



−N_high F_v/F_m





WT m

WT m



-HVFN1-mCherry-FLAG 74k Da